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SLOT-C

ECA31032- NANO ELECTRONICS

EXPERIMENT 5 - Band Structure Simulation for Graphene and CNTs using NanoHub

OUTPUT:

Assembly of Basic Applications for Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images

Bandstructure Lab - Bulk, quantum wells, nanowires

Homework Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: Si

Alloy Model: Virtual Crystal Approximation

Molar fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

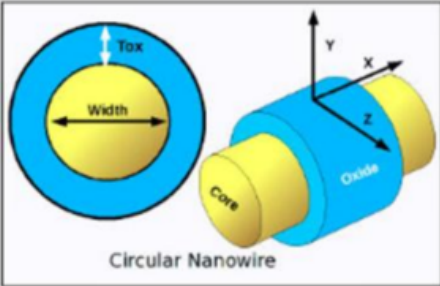
Diameter: 2.1nm

Oxide Thickness(T_{ox}): 2.2nm

Crystal Orientation

Transport direction (X): (100)

Confinement direction (Z): [010]



Circular Nanowire

Physics >

Assembly of Basic Applications for Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images

Bandstructure Lab - Bulk, quantum wells, nanowires

Homework Bandstructure Lab x Band Structure Lab

1 Device Type → 2 Physics → 3 K-Space → 4 Numerics → 5 Simulate

Electronic Structure | Strain

Tight Binding Model: sp3d5s*

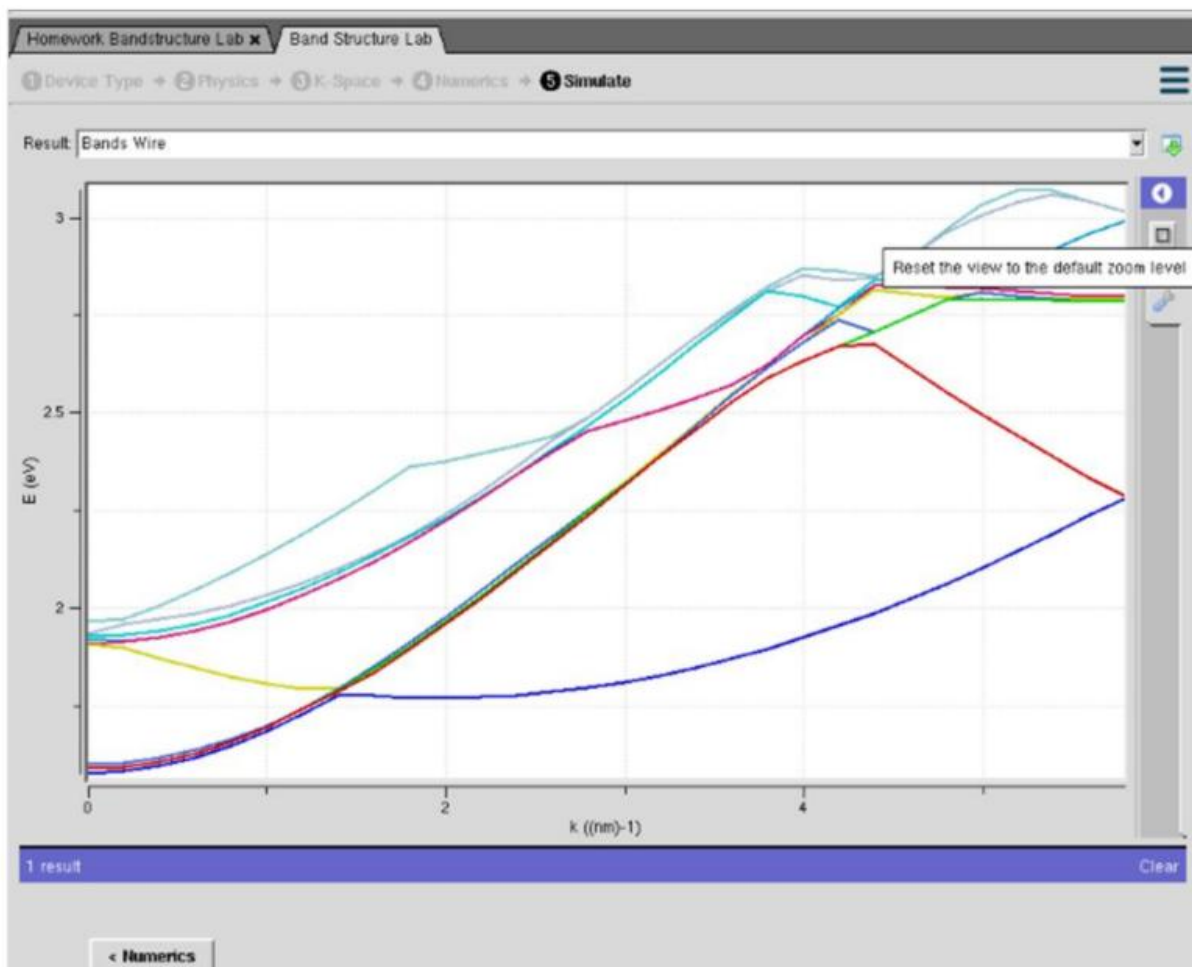
Spin-Orbit Coupling: ☒ yes ☐ no

Dangling Bond Energy: 30eV

Explore: Conduction bands

< Device Type

K-Space >



CASE STUDY 1: Simulate the band-structure of InAs circular nanowire

The simulation of the InAs (Indium Arsenide) circular nanowire is used to study its electronic band structure and energy levels. A circular nanowire model is created by defining the material properties of InAs. Due to quantum confinement, the energy levels become discrete, influencing the band gap and electron transport. The simulation generates an Energy (E) vs. Wave Vector (k) graph that shows the conduction band, valence band, and band gap. These results help in the design of high-speed transistors, infrared detectors, sensors, and other nanoelectronic devices.

Assembly of Basic Applications for Coordinated Understanding of Semiconductors

Choose Applications and Tools in this Dropdown Menu or from Images

Bandstructure Lab - Bulk, quantum wells, nanowires

Homework, Bandstructure Lab x Band Structure Lab

1 Device Type + 2 Physics + 3 K-Space + 4 Parameters + 5 Simulate

Geometry: Nanowire (1D-periodic)

Calculation For: Electrons

Material: InAs

Alloy Model: Virtual Crystal Approximation

Model fraction: 0.5

Job Type: Calculate the wire band structure

Device cross section: Circle

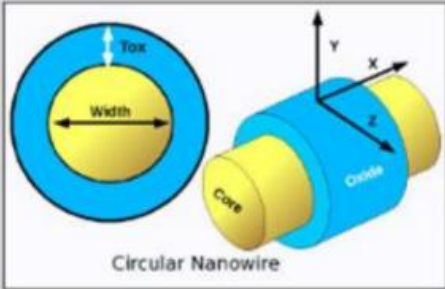
Diameter: 2.1nm

Oxide Thickness(Thickness): 2.2nm

Crystal Orientation

Transport direction (X): (100)

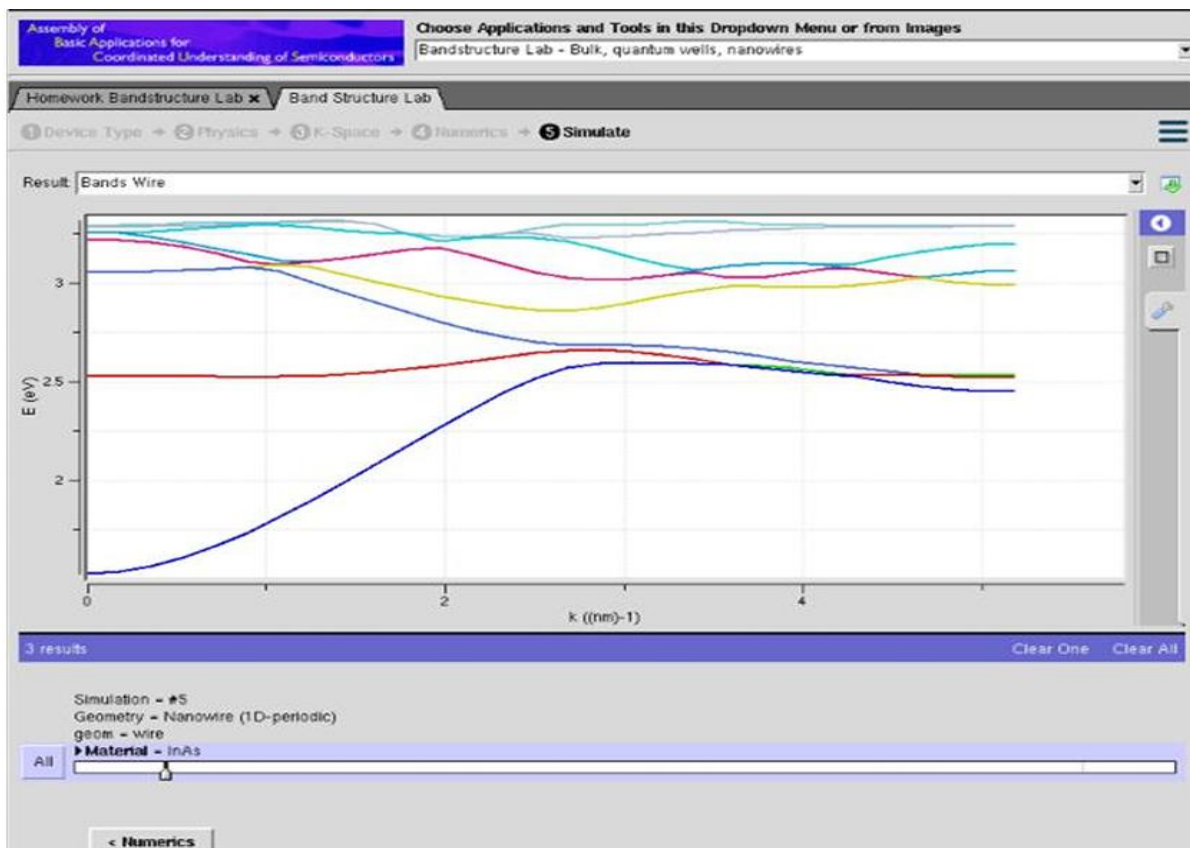
Confinement direction (Z): [010]



The diagram illustrates the geometry of a circular nanowire. On the left, a cross-sectional view shows a central yellow circle labeled 'Core' with a diameter of 2.1 nm, surrounded by a blue ring labeled 'Oxide' with a thickness of 2.2 nm. The width of the core is indicated. On the right, a 3D perspective view shows the cylindrical nanowire with a coordinate system (x, y, z) where the x-axis is along the transport direction and the z-axis is along the confinement direction. The labels 'Core' and 'Oxide' are also present in the 3D view.

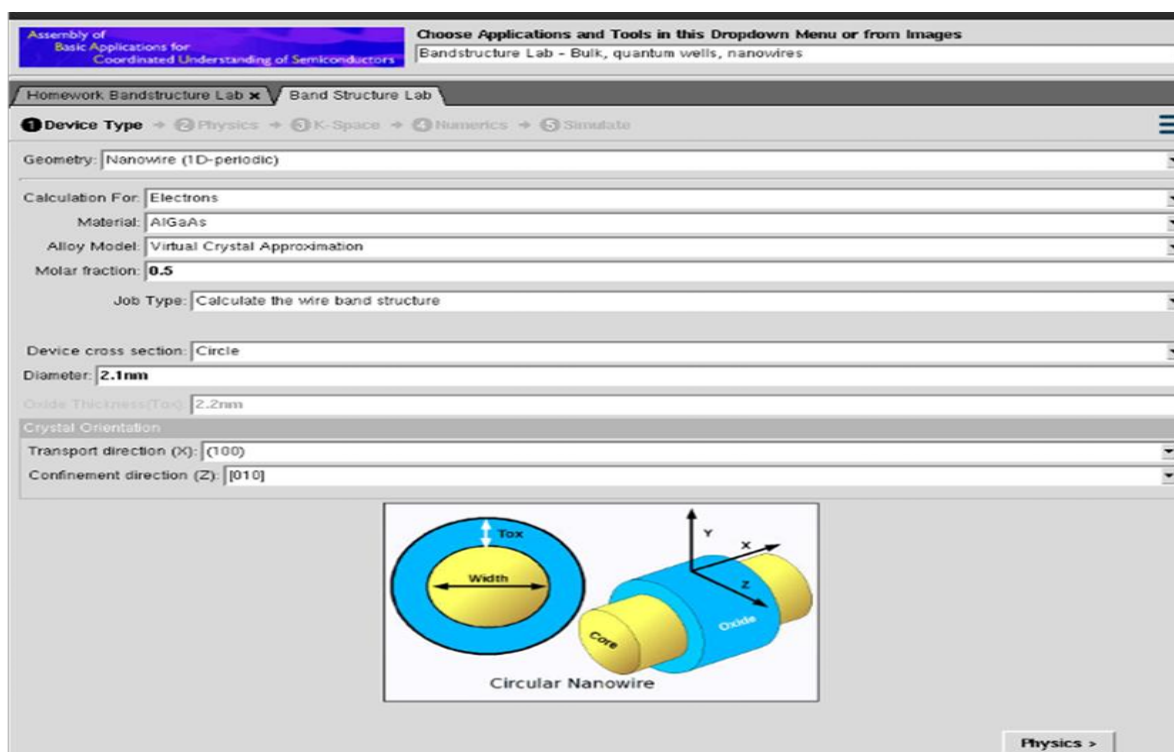
Circular Nanowire

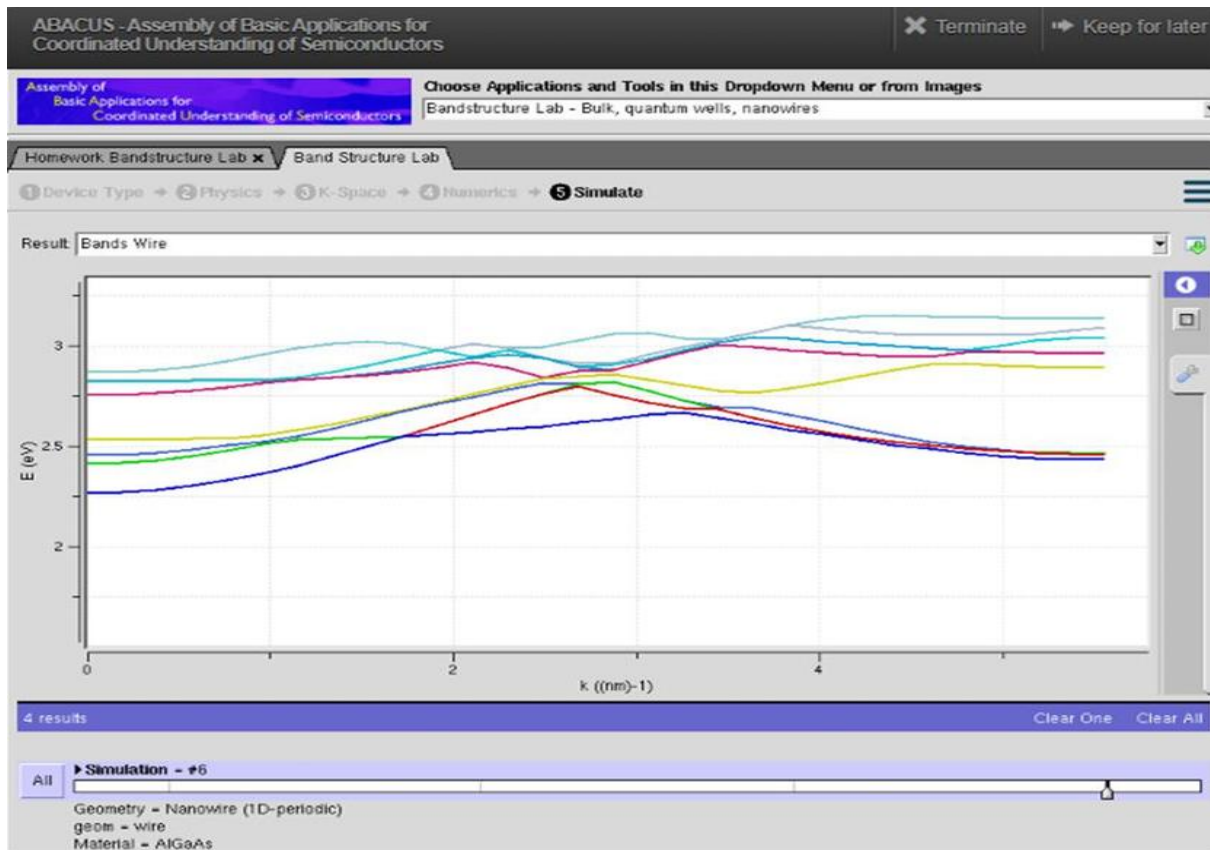
Physics >



CASE STUDY 2: Simulation of the Band Structure of AlGaAs Circular

The simulation of the AlGaAs circular nanowire is used to analyze its electronic energy levels and band gap. A circular nanowire model is created, and the material properties of AlGaAs are defined. Due to quantum confinement, the energy levels become discrete, affecting the band structure. The simulation generates an Energy (E) vs. Wave Vector (k) graph, showing the conduction band, valence band, and band gap. These results help in designing high-performance LEDs, laser diodes, photodetectors, and other nanoelectronic devices.





CONCLUSION:

The simulation of the band structure of AlGaAs and InAs circular nanowires successfully demonstrates the effect of quantum confinement on their electronic properties. The Energy (E) vs. Wave Vector (k) plots provide information about the conduction band, valence band, and band gap, helping to analyze electron behavior within the nanowires. These simulations are valuable for optimizing material properties and nanowire dimensions for high-performance nanoelectronic and optoelectronic applications, including high-speed transistors, LEDs, laser diodes, photodetectors, infrared sensors, and quantum devices.